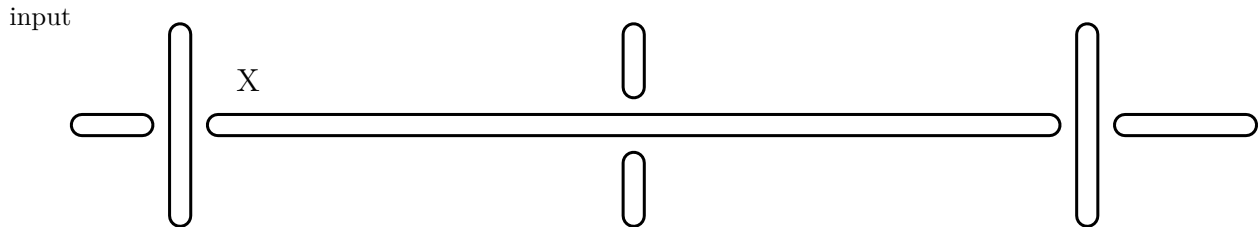
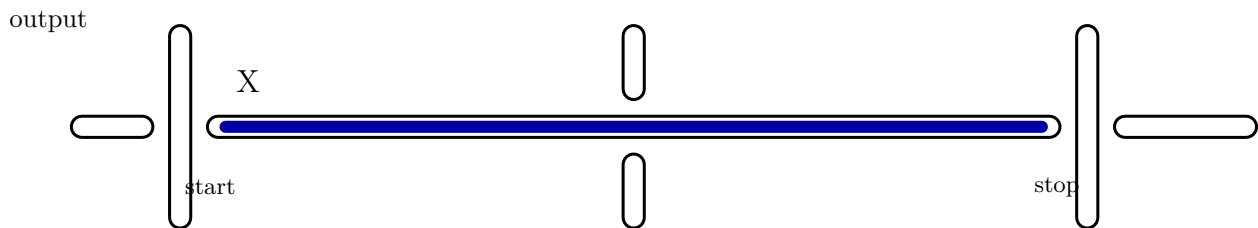


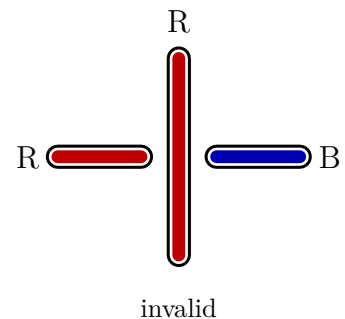
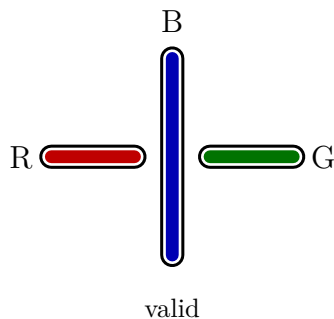
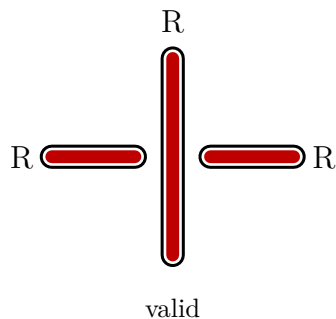
Keep cords on the table. Move them without cutting or passing strands through each other. Use red (R), blue (B), and green (G).



process: follow X across the overpass to the next gap.

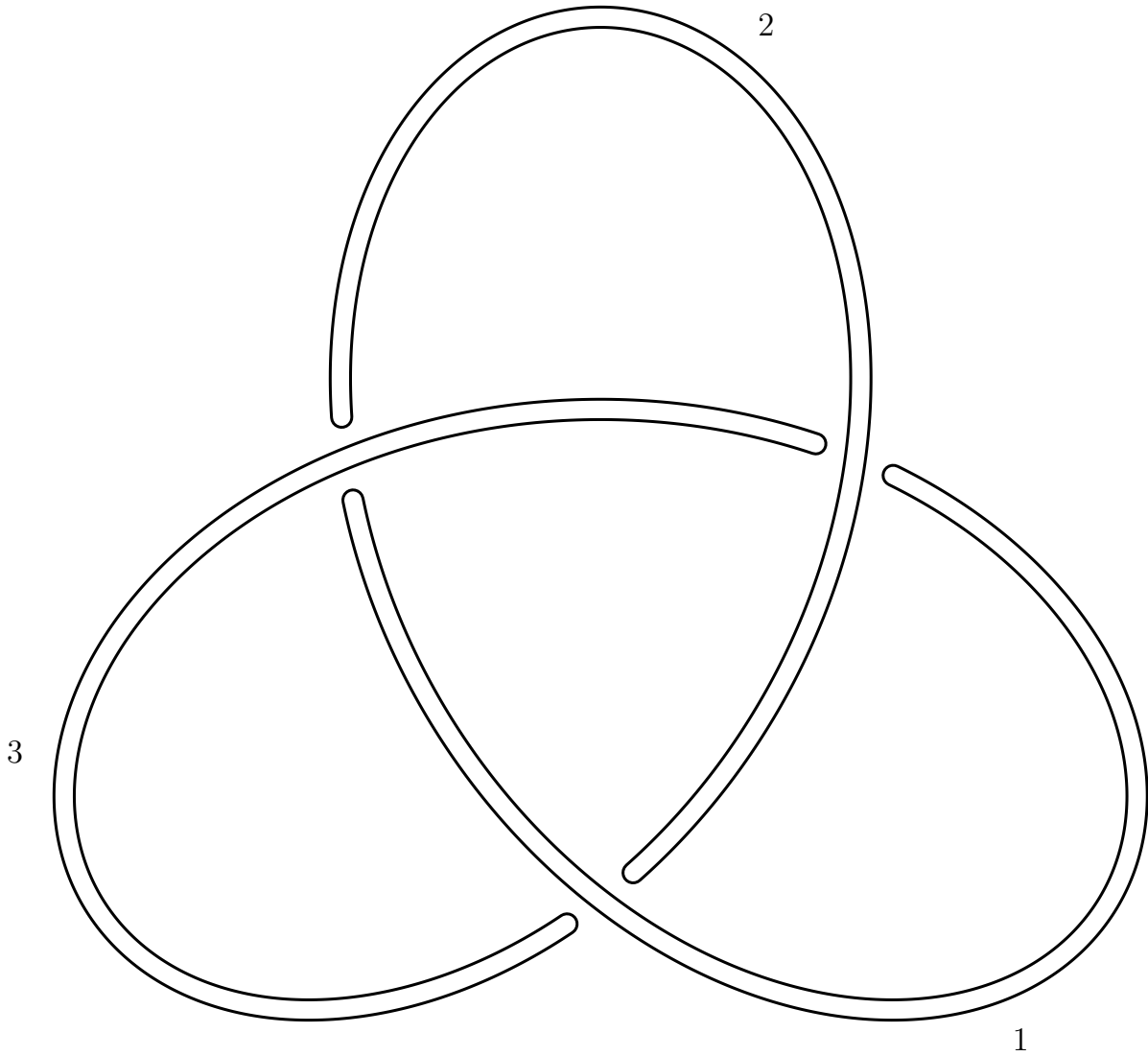


An arc runs between underpasses and continues through overpasses. A closed strand with no underpasses is one whole arc. Give each whole arc one color. At each crossing the colors must be all the same or all different.



Problem 1: Show an adult which strand goes over at each crossing above. Trace one whole arc on the next page without changing strands.

Problem 2: Find every valid coloring of this loop, including one-color choices. Record color dots for whole arcs 1, 2, and 3; use tracing paper for new tries.



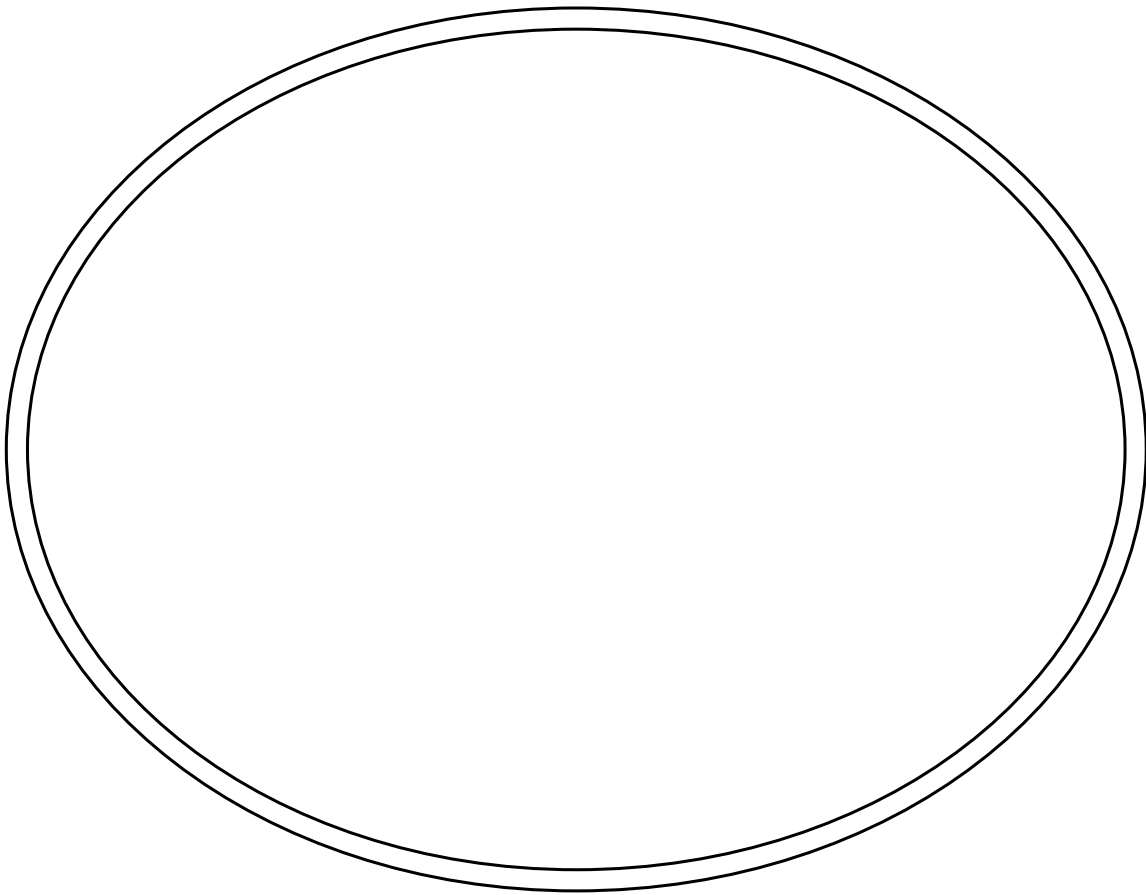
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐

1	2	3
☐	☐	☐
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐

1	2	3
☐	☐	☐
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐

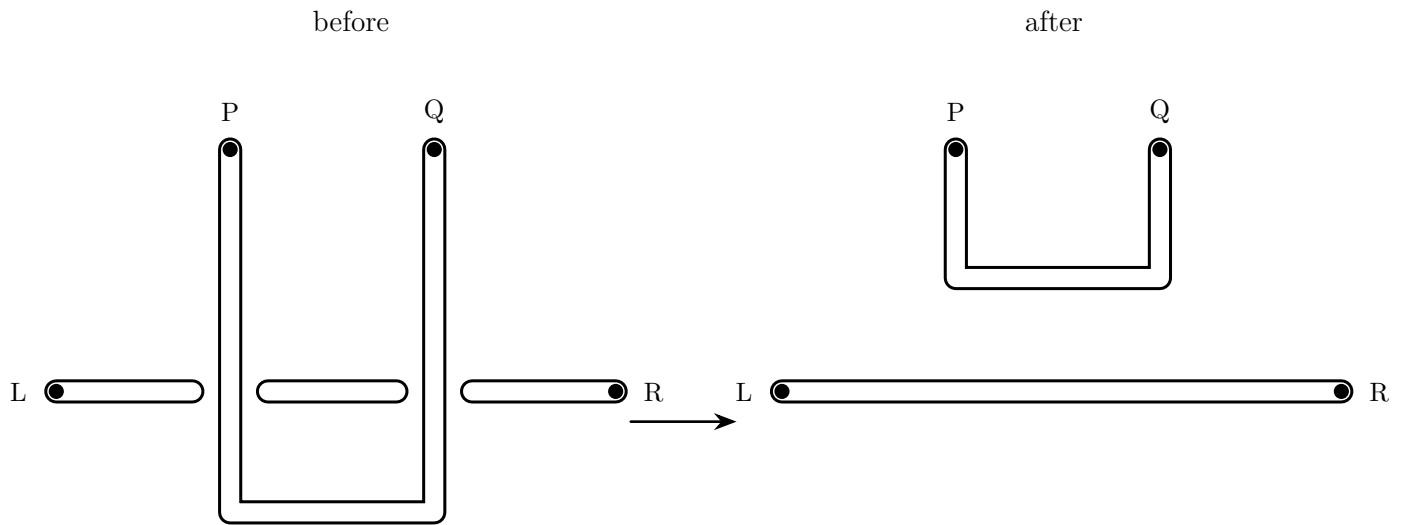
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐
1	2	3
☐	☐	☐

Problem 3: Find every valid coloring of this plain loop. Can any use more than one color?



Problem 4: Try to move each of the four adult-prepared closed cords in the tray into this plain loop shape. What can your attempts tell you, and what remains uncertain?

Problem 5: Color both pictures so matching end dots have the same colors. An arc may stop at an end dot here. Find every pair of colors at L and P that works.



Problem 6: Can the same colors at L and P give two different valid colorings of either whole picture? Explain using the crossing rule.